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Real and finite field versions of Chebotarëv's theorem

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Abstract

Chebotarev's theorem on roots of unity states that all minors of the Fourier matrix of prime size are non-vanishing. This result has been rediscovered several times and proved via different techniques. We follow the proof of Evans and Isaacs and generalize the original result to a real version and a version over finite fields. For the latter, we are able to remove an order condition between the characteristic of the field and the size of the matrix as well as decrease a sufficient lower bound on the characteristic by Zhang considerably. Direct applications include a specific real phase retrieval problem as well as a recent result for Riesz bases of exponentials.

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1 Introduction

Let $p \in \mathbb{N}$ be a prime and $\omega_p \in \mathbb{K}$ be a primitive p-th root of unity for an arbitrary field $\mathbb{K}$, then we consider the Fourier matrix

$$F_p = \left(\omega_p^{jm}\right)_{j,m=0,\dots,p-1} = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega_p & \omega_p^2 & \dots & \omega_p^{p-1} \\ 1 & \omega_p^2 & \omega_p^4 & \dots & \omega_p^{p-2} \\ 1 & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_p^{p-1} & \omega_p^{p-2} & \dots & \omega_p \end{pmatrix} \in \mathbb{K}^{p\times p},$$

and ask under which conditions all of its square submatrices are non-singular. Over the complex number field, the well known theorem by Chebotarëv gives an affirmative answer.

Theorem 1 (See e.g. [8, 15, 9, 16]; see Section 2)**.** *Let $p \in \mathbb{N}$ be prime and $\omega_p = \exp(2\pi i/p) \in \mathbb{C}$ be a p-th root of unity, then all minors of the Fourier matrix*

$$F_p \in \mathbb{C}^{p\times p}$$

are nonzero.

We will generalize this theorem to a real version and a version over finite fields as follows:

Theorem 2 (See Section 3)**.** *Let $p \in \mathbb{N}$ be prime and $\omega_p = \exp(2\pi i/p) \in \mathbb{C}$ be a p-th root of unity, then all minors of the real Vandermonde matrix*

$$\left(\left(\omega_p^j + \omega_p^{-j}\right)^m\right)_{j,m=1,\dots,\frac{p-1}{2}} \in \mathbb{R}^{\frac{p-1}{2}\times\frac{p-1}{2}}$$

are nonzero.

Theorem 3 (Informal version; see Section 4)**.** *Let $p, q \in \mathbb{N}$ be two distinct primes, $\omega_p \in \mathbb{F}_{q^{p-1}}$ be a p-th root of unity, and q sufficiently large, then all minors of the Fourier matrix*

$$F_p \in \mathbb{F}_{q^{p-1}}^{p\times p}$$

are nonzero.

The matrix in Theorem 2 is a real Vandermonde matrix and the result follows by understanding the Galois group of the field extension $\mathbb{Q} \subseteq \mathbb{Q}(\omega_p+\omega_p^{-1}) \subseteq \mathbb{Q}(\omega_p)$. Further note that Vandermonde matrices are known to be totally positive for positive nodes, see e.g. [14] or Lemma 4, and there is also a variant for a discrete cosine transform matrix which easily follows directly from Theorem 1.

Regarding Theorem 3, we note the following.

i) In [11] it has been falsely stated that the criterion $\mathrm{ord}_p(q) = p-1$ is a sufficient condition (without any lower bound on q); Zhang [17, Example 4.1,4.2] gave counterexamples for $p = 11, q = 2$ and $p = 41, q = 11$;

ii) Zhang [17, Theorem A] proved Theorem 3 under the conditions $\mathrm{ord}_p(q) = p-1$ and $q > \Gamma$, for some large Γ;

iii) Here, we get rid of the criterion $\mathrm{ord}_p(q) = p-1$ and decrease the bound Γ considerably.

The improvement on the order condition has also simultaneously and independently been done by Pfander, Caragea, Lee, and Malikiosis [4].

Beyond scientific curiosity, our results are motivated by the following applications: The real versions in Theorem 7 and Remark 9 are of interest if one wishes to do computations avoiding complex arithmetic. In particular, the complex Fourier matrix is *an* eigenvector matrix of the real symmetric graph Laplacian of the ring graph, whereas the real eigenvectors consisting of sine/cosine-pairs are the more natural choice. For the analysis of sparse signals on (more general) graphs, we refer to [6, 7]. A second application is in real phase retrieval for sparse polynomials in monomial or Chebyshev basis. The finite field version yields an uncertainty principle for specific integer matrices via standard arguments, see e.g. [16, Proof of Thm. 1.1] and Example 19. More surprisingly, a recent question in time-frequency analysis on Riesz bases of exponentials studies the principal minors of the complex Fourier matrix of size pq and reduces this to the nonvanishing minors property of the Fourier matrix of size p in characteristic q, see [3, 12]. Consequently, Corollary 17 also weakens the conditions of these results.

This paper is organized as follows: In Section 2, we recall the proof Chebotarëv's theorem given by Evans and Isaacs [8] and slightly expand it in a way, such that it can be generalized directly to a real version in Section 3. For this proof Schur polynomials are essential. Now, the known proofs always use a divisibility argument, that is valid in $\mathbb{Z}$ but not in characteristic $q > 0$. Thus a more delicate analysis is required in Section 4. From this analysis we also obtain a new proof of Theorem 1 that completely works in the field $\mathbb{Q}(\omega_p)$ and does not rely on a substitution $\omega_p \mapsto 1$, which is a standard trick in the known proofs, see Remark 14.

2 Prerequisites

An important tool for our proofs are Schur polynomials. For $k \in \mathbb{N}$, a set $A = \{a_1, \dots, a_k\} \in \binom{\{0,1,\dots,n-1\}}{k}$, with $a_1 < a_2 < \ldots < a_k$, and any field $\mathbb{K}$ consider the generalized Vandermonde matrix

$$V_A = V_A(x_1, \dots, x_k) = \begin{pmatrix} x_1^{a_1} & x_2^{a_1} & x_3^{a_1} & \dots & x_k^{a_1} \\ x_1^{a_2} & x_2^{a_2} & x_3^{a_2} & \dots & x_k^{a_2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1^{a_k} & x_2^{a_k} & x_3^{a_k} & \dots & x_k^{a_k} \end{pmatrix} \in (\mathbb{K}[x_1, \dots, x_k])^{k \times k}, \tag{2.1}$$

and drop the subscript for $A = \{0, \dots, k-1\}$. For any matrix M, denote by $M_{A,B}$ the restriction of M to the rows A and the columns B. With this notation we observe $V_A(x_1, \dots, x_k) = (V_{\{0,\dots,n-1\}}(x_1, \dots, x_n))_{A,\{1,\dots,k\}}$.

The determinant $\det(V_A)$ vanishes if $(x_i = x_j)$ and thus $(x_j - x_i) | \det(V_A)$. This motivates the definition of the Schur polynomial $s_A(x_1, \dots, x_k)$ as

$$s_A(x_1, \dots, x_k) = \frac{\det(V_A)}{\prod_{1 \le i < j \le k} (x_j - x_i)}.$$

The following results is due to Mitchell [13], but is stated in a modern way in [8].

Lemma 4 (See [13, 8]). *Let $A \in \binom{\{0,1,\dots,n-1\}}{k}$ be any index set. The Schur polynomial $s_A(x_1,\dots,x_k) \in \mathbb{Q}[x_1,\dots,x_k]$ has nonnegative coefficients and*

$$s_A(1,\dots,1) = \frac{\prod\limits_{1\le i<j\le k}(a_j - a_i)}{\prod\limits_{1\le i<j\le k}(j-i)}.$$

These tools are already sufficient to prove Theorem 1. We will provide a proof that might seem cumbersome at some points, but can directly be generalized to a real version.

Proof of Theorem 1. Let $A, B \in \binom{\{0,1,\dots,p-1\}}{k}$ be the indices of a submatrix

$$M = \begin{pmatrix} \omega_p^{a_1 b_1} & \omega_p^{a_1 b_2} & \dots & \omega_p^{a_1 b_k} \\ \omega_p^{a_2 b_1} & \omega_p^{a_2 b_2} & \dots & \omega_p^{a_2 b_k} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_p^{a_k b_1} & \omega_p^{a_k b_2} & \dots & \omega_p^{a_k b_k} \end{pmatrix}$$

of the Fourier matrix F_p, with $\omega_p = \exp(2\pi i/p)$. We consider the Schur polynomial $s_A(x_1,\dots,x_k)$ and observe that $\det(M) = 0$ if and only if $s_A(\omega_p^{b_1}, \omega_p^{b_2}, \dots, \omega_p^{b_k}) = 0$. After the substitution

$$\alpha_B \colon \mathbb{Q}[x_1,\dots,x_k] \to \mathbb{Q}[X],\ x_l \mapsto X^{b_l},$$

the multivariate Schur polynomial s_A becomes an univariate polynomial $\alpha_B(s_A) \in \mathbb{Q}[X]$. The prior statement can be rewritten as $\alpha_B(s_A)(\omega_p) = 0$ if and only if $\det(M) = 0$. Now, since $\mathbb{Q}[X]$ is a principal ideal domain, $\alpha_B(s_A)(\omega_p) = 0$ implies that $\alpha_B(s_A)$ is divisible by the minimal polynomial $\Phi_p(X) = \sum_{i=0}^{p-1} X^i$ of ω_p, i.e.,

$$\alpha_B(s_A) = \Phi_p(X) \cdot R(X), \tag{2.2}$$

for some $R \in \mathbb{Q}[X]$. Both polynomials, $\alpha_B(s_A)$ and Φ_p, have integer coefficients and leading coefficient 1. Hence the same holds for R. Equality (2.2) also holds true after another substitution $X \mapsto 1$ and yields

$$\alpha_B(s_A)(1) = \underbrace{\Phi_p(1)}_{=p} \cdot R(1). \tag{2.3}$$

Finally, we want to show that $\alpha_B(s_A)(1)$ is not divisible by the prime p, which contradicts $\det(M) = 0$. Here we can use that the value $X = 1$ has good properties under the substitution $x_l \mapsto X^{b_l}$, i.e.,

$$s_A(1,1,\dots,1) = \alpha_B(s_A)(1) = s_A(1,1,\dots,1) = \frac{\prod\limits_{1\le i<j\le k}(a_j - a_i)}{\prod\limits_{1\le i<j\le k}(j-i)},$$

by Lemma 4. The factors $(a_j - a_i)$ are all smaller than p and cannot divide p. This contradicts equation (2.3) and hence $\det(M) \neq 0$. ∎

A key ingredient in the proof is that there exists a substitution $\alpha_B \colon \mathbb{Q}[x_1,\dots,x_k] \to \mathbb{Q}[X]$ with $\alpha(s_A) \in \mathbb{Q}[X]$ and $s_A(\omega_p^{b_1},\dots,\omega_p^{b_k}) = \alpha_B(s_A)(\omega_p)$. This substitution only exists because of the polynomial connection between the roots $\omega_p^{b_i}$ that is encoded in the fact that $\mathbb{Q}(\omega_p) = \mathbb{Q}[X]/\Phi_p(X)$ is the splitting field of $\Phi_p(X)$. In this case we have $\mathrm{Gal}(\mathbb{Q}[X]/\Phi_p(X)|\mathbb{Q}) = (\mathbb{Z}/p)^\times$. Furthermore, we were able to evaluate at the point $x_1 = \dots = x_k = X = 1$, here it was crucial that 1 is a fixed point of the maps $x_l \mapsto X^{b_l}$ and the diagram in Figure 2.1 commutes.

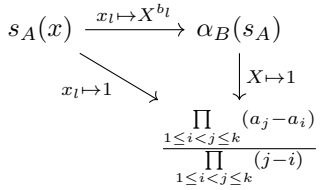

Figure 2.1: The Substitutions of the Schur Polynomial in the proof of Theorem 1.

3 Real version of Chebotarëv's theorem

For each prime $p > 3$ the group $(\mathbb{Z}/p)^\times$ is the Galois group of the field extension $\mathbb{Q} \subseteq \mathbb{Q}[X]/\Phi_p(X)$ and has the subgroup $H = \{1, -1\}$. This subgroup H is generated by the automorphism on $\mathbb{Q}[X]/\Phi_p(X)$ which is induced by the complex conjugation $i \mapsto -i$. By the Galois correspondence this induces the subfield

$$\mathbb{Q} \subsetneq \mathrm{Fix}(H) \subsetneq \mathbb{Q}[X]/\Phi_p(X),$$

of degree $\deg_{\mathbb{Q}}(\mathrm{Fix}(H)) = \frac{p-1}{2}$. This field is generated as $\mathbb{Q}(\omega_p + \omega_p^{-1}) = \mathrm{Fix}(H) \subseteq \mathbb{R}$ and has the cyclic Galois group

$$\mathbb{Z}_p^\times / H \cong \mathbb{Z}_{\frac{p-1}{2}},$$

which is generated by the image of any generator $(\mathbb{Z}/p)^\times$ modulo H. It is $\omega_p + \omega_p^{-1} = 2\cos(2\pi/p)$ and these elements are mapped to $(\omega_p^j + \omega_p^{-j}) = 2\cos(2\pi j/p)$ by the Galois group of the cyclotomic field. Thus the elements

$$(\omega_p^j + \omega_p^{-j}) = 2\cos(2\pi j/p), \quad j = 1, \ldots, \frac{p-1}{2},$$

are conjugate and share the same minimal polynomial

$$P(X) = \prod_{j=1}^{(p-1)/2} (X - 2\cos(2j\pi/p)).$$

The coefficients of this polynomial are symmetric in the cosine terms and hence, due to the Galois correspondence, must lie in the fixed field of the Galois group, i.e., the coefficients are rational. An explicit formula for the coefficients of this minimal polynomial is given in [5] by

$$P(x) = \sum_{0 \le k \le \frac{n}{2}} (-1)^k \binom{n-k}{k} X^{n-2k} + \sum_{0 \le k < \frac{n}{2}} (-1)^k \binom{n-k-1}{k} X^{n-2k-1},$$

where $n = \frac{p-1}{2}$. However, this expression does not provide much additional insight into the structure of the minimal polynomial and will not be used later. In order to use the Schur polynomials, we need to understand the polynomial expression between the roots.

Lemma 5. *Let p be a prime and let $P(X) \in \mathbb{Q}[X]$ be the minimal polynomial of $2\cos(2\pi/p)$. The elements $\varphi_j \in \mathbb{Q}[X]/P(X)$ that are given by the recursion*

$$\begin{aligned} &\varphi_1(X) = X, \\ &\varphi_2(X) = X^2 - 2, \\ &\varphi_j(X) = X \cdot \varphi_{j-1}(X) - \varphi_{j-2}(X), \qquad j = 3, \ldots, \frac{p-1}{2}, \end{aligned}$$

are to the roots of P. Viewed over the rationals, the polynomials $\varphi_j(X)$ have the common fixed point at 2, i.e., $\varphi_j(2) = 2$ for all j.

Proof. In the field $\mathbb{Q}[X]/P(X)$ we can use $X = (\omega_p + \omega_p^{-1})$ and we have

$$X^2 - 2 = \omega_p^2 + \omega_p^{-2}.$$

Induction on j proves the first part, where we use

$$\varphi_j(X) = (\omega_p + \omega_p^{-1}) \cdot (\omega_p^{j-1} + \omega_p^{1-j}) - (\omega_p^{j-2} + \omega_p^{2-j}) = \omega_p^j + \omega_p^{-j}.$$

Regarding the second part, we see that $\varphi_j(2) = 2 \cdot \varphi_{j-1}(2) - \varphi_{j-2}(2) = 2 \cdot 2 - 2 = 2$. ∎

Now, we evaluate the minimal polynomial $P(X)$ at the fixed point 2 of the polynomials φ_j, $j = 1, \ldots, \frac{p-1}{2}$.

Lemma 6. *Let p be a prime. The minimal polynomial $P(X)$ of $2\cos(2\pi/p)$ has the evaluation $P(2) = p$.*

Proof. The Chebyshev polynomials of first kind $T_p \colon [-1, 1] \to \mathbb{R}$ are given by

$$T_p(x) = \cos(p \arccos x).$$

The roots of

$$P(X) = \prod_{j=1}^{(p-1)/2} \left(X - 2\cos\frac{2j\pi}{p} \right)$$

coincide with the local maxima of $2T_p(X/2)$ in $(-2, 2)$, $2T_p(X/2) = 2$, and $2T_p(X/2)$ is a monic polynomial of degree p. Hence we have the identity

$$2(T_p(X/2) - 1) = (P(X))^2(X - 2)$$

from which also

$$(P(2))^2 = \lim_{y \to 1} \frac{T_p(y) - 1}{y - 1} = T_p'(1) = p \lim_{\theta \to 0} \frac{\sin p\theta}{\sin \theta} = p^2$$

follows. ∎

We now have all the tools to state and prove the main theorem of this section, which can be seen as a real version of Chebatarëv's theorem.

Theorem 7. *Let p be a prime. All minors of the Vandermonde matrix*

$$\left(\left(\cos\left(\frac{2 \cdot j \cdot \pi}{p} \right) \right)^m \right)_{j,m=1,\ldots,\frac{p-1}{2}} \in \mathbb{R}^{\frac{p-1}{2} \times \frac{p-1}{2}}$$

are nonzero.

Proof. For notational convenience, we work with the Vandermonde matrix, see Equation (2.1),

$$V\left(2 \cdot \cos\left(\frac{2 \cdot j \cdot \pi}{p} \right)_{j=1,\ldots,\frac{p-1}{2}} \right),$$

whose columns are a multiple of the original matrix and hence the minors show the same vanishing behaviour. Let $A, B \in \binom{[\frac{p-1}{2}]}{l}$ be the indices of the submatrix that is under consideration. This minor is zero if and only if $s_A(2\cos(2b_1\pi/p), \ldots, 2\cos(2b_l\pi/p)) = 0$. We break down the step from $s_A(x_1, \ldots, x_k)$ to $s_A(2\cos(2b_1\pi/p), \ldots, 2\cos(2b_l\pi/p)) = 0$ in two steps. The first step is the substitution

$$\alpha \colon \mathbb{Q}[x_1, \ldots, x_l] \to \mathbb{Q}[X],\ x_i \mapsto \varphi_{b_i}(X),$$

with φ_{b_i} from Lemma 5. The second step is to look at the image of $\alpha_B(s_A)$ modulo the minimal polynomial P of $2\cos(2\pi/p)$. The minor is zero if and only if $P|\alpha_B(s_A)$ in $\mathbb{Q}[X]$. If this equation holds, it will also hold after another substitution $X \mapsto 2$. Lemma 6 yields $P(2) = p$ and Lemma 4 yields

$$\begin{aligned} \alpha_B(s_A)(2,\dots,2) &= s_A(2,\dots,2) && (3.1)\\ &= 2^d \cdot s_A(1,\dots,1) && (3.2)\\ &= 2^d \cdot \frac{\prod\limits_{1\le i<j\le l}(a_j - a_i)}{\prod\limits_{1\le i<j\le l}(j-i)}, && (3.3) \end{aligned}$$

which is a contradiction to the divisibility by p, hence the minor must be nonzero. Equality (3.1) follows from the fact that 2 is a fixed point of all φ_k, equality (3.2) follows from the fact that Schur polynomials are homogeneous of some degree d and equality (3.3) is Lemma 4. ■

Example 8. *Let $p = 7$ and $t_1 = \cos(2\pi/7) \approx 0.62$, $t_2 = \cos(4\pi/7) \approx -0.22$, and $t_3 = \cos(6\pi/7) \approx -0.90$ be given. Then all minors of the Vandermonde matrix*

$$\begin{pmatrix} 1 & t_1 & t_2^2 \\ 1 & t_2 & t_2^2 \\ 1 & t_3 & t_3^2 \end{pmatrix}$$

are nonzero, while (obviously) the matrix is not totally positive.

Remark 9. *Another real version can be infered from the classical Chebotarëv result. Let p be a prime, set $r = (p-1)/2$, and*

$$F = \left(\exp\frac{2\pi \mathrm{i} kj}{p}\right)_{k,j=-r,\dots,r} \in \mathbb{C}^{p\times p}, \qquad C = \left(2\cos\frac{2\pi kj}{p}\right)_{k,j=1,\dots,r} \in \mathbb{C}^{r\times r}.$$

Now assume $\det(C_{J\times K}) = 0$ for some $J, K \in \binom{[r]}{s}$. Then there exists a $\hat{f} = (\hat{f}_{-r},\dots,\hat{f}_r)^\top \in \mathbb{C}^p$, $\hat{f}_k = \hat{f}_{-k}$, which fulfils

$$0 = \sum_{k\in K} 2\hat{f}_k \cos\frac{2\pi kj}{p} = \sum_{k\in K\cup -K} \hat{f}_k \exp\frac{2\pi \mathrm{i} kj}{p}$$

for $j \in J$, as well as for $j \in -J$ in the first equation since the cosine is even. This implies $\det(F_{(J\cup -J)\times(K\cup -K)}) = 0$ which is false.

More general results on uncertainty principles for vectors with symmetry and related Chebotarëv theorems can be found in [10].

Finally, we note in passing that polynomials that are either sparse in the monomial basis (using Theorem 7) or the Chebyshev basis (using Remark 9) are uniquely determined by a minimal number of *phaseless* evaluations. This is a direct consequence since the non-vanishing minors property implies the so-called complement property, which is equivalent to the uniqueness of the real phase-retrieval problem, see e.g. [1, 2].

4 Chebotarëv's theorem over finite fields

Now let us once again consider the Fourier matrix F_p of prime size, but this time over fields of characteristic q for $q \in \mathbb{N}$ prime. It is well known that, given two primes $p, q \in \mathbb{Z}$, the polynomial

$$\overline{\Phi_p(X)} = \sum_{i=0}^{p-1} X^i \in \mathbb{F}_q[X]$$

is not even necessarily irreducible. In fact $\overline{\Phi_p(X)}$ is irreducible if and only if $\mathrm{ord}_p(q) = p-1$. This follows from the fact that, if Φ_p is irreducible, the roots of $\overline{\Phi_p(X)}$ are given by $X^{q^i} \in \mathbb{F}_q[X]/\Phi_(X)$, for $i = 0, 1, \ldots, p-2$, and the roots fulfill $X^p = 1$. Thus the roots are given by

$$X, X^{(q \mod p)}, X^{(q^2 \mod p)}, X^{(q^3 \mod p)}, \ldots, X^{(q^{p-2} \mod p)},$$

and these are repeated, if $\mathrm{ord}_p(q) = r < p-1$, which would contradict the separability of $\overline{\Phi_p(X)}$. In fact it is known that $\overline{\Phi_p(X)} \in \mathbb{F}_q[X]$ splits into $\frac{p-1}{r}$ many irreducible polynomials of degree r. The roots of all all these polynomials are contained in the field $\mathbb{F}_{q^r}$.

Example 10. *For $p = 7$ and $q = 11$ we have $r = \mathrm{ord}_7(11) = 3$ and the factorization*

$$(X^3 + 7X^2 + 6X + 10)(X^3 + 5X^2 + 4X + 10) = (X^6 + X^5 + X^4 + X^3 + X^2 + X + 1) \in \mathbb{F}_{11}[X].$$

Zhang's conditions [17], $\mathrm{ord}_p(q) = p-1$ and $q > \Gamma$, for some large Γ, imply the nonvanishing of the minors, which on polynomial side again can be written as

$$\overline{\Phi_p(X)} \nmid \alpha_B(s_A(x_1, \ldots, x_k)) \in \mathbb{F}_q[X],$$

for all $A, B \in \binom{\{0,1,\ldots,p-1\}}{k}$. The sufficient lower bound is given by

$$\Gamma = \max_{A \subseteq \{0,1,\ldots,p-1\}} \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)},$$

and for $q > \Gamma$, the Euclidean division

$$\alpha_B(s_A(x_1, \ldots, x_k)) = f(X) \cdot \Phi_p(X) + R(X) \in \mathbb{Z}[X]$$

in combination with

$$\alpha_B(s_A(x_1, \ldots, x_k))(1) = \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)},$$

yields that q is too large to divide all the coefficients of $R(X)$ and thus $\overline{R(X)} \neq 0 \in \mathbb{F}_q[X]$. We want to improve the condition on q with an approach that provides more insight into the problem. From now on, let p, q be two distinct primes with $\mathrm{ord}_p(q) = r$ and denote $\Phi_p(X) = \sum_{i=0}^{p-1} X^i \in \mathbb{Z}[X]$ and $\overline{\Phi_p(X)} = \sum_{i=0}^{p-1} X^i \in \mathbb{F}_q[X]$. Since $\overline{\Phi_p(X)}$ is not necessarily irreducible, we denote by $\overline{P(X)} \in \mathbb{F}_q[X]$ an irreducible factor of $\overline{\Phi_p(X)}$ and by $P(X) \in \mathbb{Z}[X]$ the canonical preimage of $\overline{P(X)}$ in $\mathbb{Z}[X]$. Note that all the nontrivial $(p-1)$ many p-th roots of unity ω_p will be contained in the field $\mathbb{F}_{q^r} \cong \mathbb{F}_q[X]/\overline{P(X)}$, with one explicit root given by $\omega_p = X$. Let

$$F_p = V(1, \omega_p, \omega_p^2, \omega_p^3, \ldots, \omega_p^{p-1}) \in (\mathbb{F}_{q^r})^{p \times p}. \tag{4.1}$$

be the Fourier matrix of size p in characteristic q. This matrix always has entries in $\mathbb{F}_{q^r}$.

Example 11. *Again, for $p = 7$ and $q = 11$ we know that $\mathbb{F}_{11^3} \cong \mathbb{F}_{11}[X]/(X^3 + 7X^2 + 6X + 10)$ is the splitting field of $\overline{\Phi_p(X)}$ and one explicit 7-th root of unity is $\omega_7 = X$. The Fourier matrix is given by*

$$F_7 = V(1, X, X^2, X^3, X^4, X^5, X^6) \in (\mathbb{F}_{11}[X]/(X^3 + 7X^2 + 6X + 10))^{7 \times 7},$$

with $X^3 = 4X^2 + 5X + 1$, $X^4 = -X^2 - X - 4$, $X^5 = -5X^2 - X - 1$ and $X^6 = X^2 - 4X - 5$ in the field $\mathbb{F}_{11^3} = \mathbb{F}_{11}[X]/(X^3 + 7X^2 - 6X + 10)$.

Furthermore, for $B \in \binom{\{0,1,\ldots,p-1\}}{k}$ and $l \in (\mathbb{F}_q)^\times$ we denote

$$l \cdot B := \{(l \cdot b) \mod p \colon b \in B\}.$$

The multiplication with l is a bijection on $\mathbb{F}_p^\times$. We start with a simple well known observation.

Lemma 12. *Let $p, q \in \mathbb{Z}$ be two distinct primes with $\mathrm{ord}_p(q) = r$. Let $F_p \in \mathbb{F}_{q^r}^{p\times p}$ be the Fourier matrix in characteristic q, see (4.1). Let $A, B \in \binom{\{0,1,\dots,p-1\}}{k}$ be two index sets. We have*

$$\det((F_p)_{A,B}) = 0$$

if and only if

$$\det((F_p)_{A,l\cdot B}) = 0$$

for any $l \in (\mathbb{F}_p)^\times$.

Proof. If $\det((F_p)_{A,B}) = 0$, there are coefficients $c_1, \dots, c_k$, such that

$$\sum_{j=1}^{k} c_j \omega_p^{(a-1)(b_j-1)} = 0$$

for all $a \in A$. Substituting B with $(l \cdot B)$ yields

$$\sum_{j=1}^{k} c_j \omega_p^{(a-1)(lb_j-1)} = \omega_p^{(a-1)((l-1)b_j)} \sum_{j=1}^{k} c_j \omega_p^{(a-1)(b_j-1)} = 0$$

for all $a \in A$. ∎

$$\begin{array}{ccc} \mathbb{Z}[X] & \xrightarrow{\bmod \Phi_p(X)} & \mathbb{Z}[X]/\Phi_p(X) \\ \Big\downarrow{\scriptstyle \bmod q} & & \Big\downarrow{\scriptstyle \bmod q} \\ \mathbb{F}_q[X] & \xrightarrow{\bmod \overline{\Phi_p(X)}} & \mathbb{F}_q[X]/\overline{\Phi_p(X)} \end{array}$$

Figure 4.1: The p-th cyclotomic polynomial over $\mathbb{Z}$ and $\mathbb{F}_q$, with $\mathrm{ord}_p(q) = p - 1$.

For $\mathrm{ord}_p(q) = p - 1$ consider the commuting diagram in Figure 4.1. We want to know whether

$$\overline{\Phi_p(X)} \mid \overline{\alpha_B(s_A(x_1, \dots, x_k))} \in \mathbb{F}_q[X]/\overline{\Phi_p(X)} \tag{4.2}$$

for some $A, B \in \binom{\{0,1,\dots,p-1\}}{k}$. We know by Theorem 1 that

$$\Phi_p(X) \nmid \alpha_B(s_A(x_1, \dots, x_k)) \in \mathbb{Z}[X]$$

for all $A, B \in \binom{\{0,1,\dots,p-1\}}{k}$. Euclidean division in $\mathbb{Z}[X]$ yields

$$\alpha_B(s_A(x_1, \dots, x_k)) = f(X) \cdot \Phi_p(X) + R(X),$$

with $R \neq 0$ and thus equation (4.2) holds if q divides all the coefficients of $R(X)$. The polynomial R itself may be quite complicated, but averaging over well chosen column sets B can be reasonably described. For two fixed sets $A, B \in \binom{\{0,1,\dots,p-1\}}{k}$ denote by $m_{A,B}$ the number of indices μ of $s_A(x_1, \dots, x_k) = \sum_{\mu\in T} c_\mu x^\mu$, such that p divides $\sum_{j=1}^{k} \mu_j b_j$, i.e.,

$$m_{A,B} = \sum_{\mu\in T} c_\mu \mathbb{1}\left(p \Big| \sum_{j=0}^{k} \mu_j b_j\right).$$

Here $b_1 < \ldots < b_k$ is the ascending list of the elements of B. By abuse of notation, we also write $\langle \mu, B\rangle = \sum_{j=1}^{k} \mu_j b_j$.

Lemma 13. *Let $p \in \mathbb{Z}$ be a prime and let $A, B \in \binom{\{0,1,\ldots,p-1\}}{k}$ be two sets. We have*

$$\sum_{l=1}^{p-1} \alpha_{(l\cdot B)}(s_A(x_1,\ldots,x_k)) = m_{A,B}\cdot p - \frac{\prod_{i<j}(a_j-a_i)}{\prod_{i<j}(j-i)} \in \mathbb{Z}[X]/\Phi_p(X).$$

Proof. The ring $\mathbb{Z}[X]/\Phi_p(X)$ is a $\mathbb{Z}$-algebra and all substitutions $\alpha_{(q^l\cdot B)}$ are $\mathbb{Z}$-linear maps. Thus we observe

$$\begin{aligned}\sum_{l=1}^{p-1} \alpha_{(l\cdot B)}(s_A(x_1,\ldots,x_k)) &= \sum_{l=1}^{p-1} \alpha_{(l\cdot B)}\left(\sum_{\mu\in T} c_\mu x^\mu\right)\\ &= \sum_{l=1}^{p-1}\left(\sum_{\mu\in T} c_\mu \alpha_{(l\cdot B)}(x^\mu)\right)\\ &= \sum_{\mu\in T} c_\mu\left(\sum_{l=1}^{p-1}\alpha_{(l\cdot B)}(x^\mu)\right).\end{aligned}$$

We distinguish two cases, first $p|\langle\mu,B\rangle$ and second $p\nmid\langle\mu,B\rangle$. In the first case, we have

$$\alpha_B(x^\mu) = \prod_{j=1}^{k} X^{\mu_j b_j} = X^{\sum_{j=1}^k \mu_j b_j} = X^{\langle\mu,B\rangle} = 1,$$

since $X^p = 1$. This yields

$$\sum_{l=1}^{p-1}\alpha_{l\cdot B}(x^\mu) = \sum_{l=1}^{p-1} 1 = p-1.$$

In the second case we have $\langle\mu,B\rangle = z \neq 0 \mod p$. For any $l \in (\mathbb{F}_p)^\times$ we observe

$$\alpha_{(l\cdot B)} = \prod_{j=1}^{k} X^{\mu_j\cdot b_j\cdot l} = X^{l\langle\mu,B\rangle} = X^{l\cdot z}$$

Now, since multiplication with z is an isomorphism on $\mathbb{F}_p^\times$, we have

$$\sum_{l=1}^{p-1} X^{l\cdot z} = \sum_{l=1}^{p-1} X^l = -1,$$

in $\mathbb{Z}[X]/\Phi_p(X)$. There are, counted with multiplicity, $\alpha_B(s_A(x_1,\ldots,x_k))(1) = \frac{\prod_{i<j}(a_j-a_i)}{\prod_{i<j}(j-i)}$ many exponents in s_A. We finally conclude

$$\begin{aligned}\sum_{\mu\in T} c_\mu\left(\sum_{l=1}^{p-1}\alpha_{(l\cdot B)}(x^\mu)\right) &= \sum_{\mu\in T:\ p|\langle\mu,B\rangle} c_\mu\left(\sum_{l=1}^{p-1}\alpha_{(l\cdot B)}(x^\mu)\right) + \sum_{\mu\in T:\ p\nmid\langle\mu,B\rangle} c_\mu\left(\sum_{l=1}^{p-1}\alpha_{(l\cdot B)}(x^\mu)\right)\\ &= m_{A,B}\cdot(p-1) - \frac{\prod_{i<j}(a_j-a_i)}{\prod_{i<j}(j-i)} + m_{A,B}\\ &= m_{A,B}\cdot p - \frac{\prod_{i<j}(a_j-a_i)}{\prod_{i<j}(j-i)}.\end{aligned}$$

■

Remark 14. *Note that the proof also works for every other arbitrary list $b_1,\ldots,b_k$ of B. So in principle, one can choose the list of B with the best properties.*

Note further that Lemma 13 yields yet another proof for Theorem 1, without an explicit evaluation at $X = 1$. Since

$$\frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)}$$

is not divisible by p in $\mathbb{Z}$, we have

$$\sum_{l=1}^{p-1} \alpha_{(l\cdot B)}(s_A(x_1, \ldots, x_k)) = m \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \neq 0$$

and thus $\alpha_B(s_A) \neq 0 \mod \Phi_p(X)$.

Since the quantity $m_{A,B} \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \in \mathbb{Z}[X]/\Phi_p(X)$ is an integer, it is easier to understand the rightmost map in Figure 4.1. For $\mathrm{ord}_p(q) = r < p-1$, the diagram in Figure 4.2 looks a bit different. There still is a map $\varphi\colon \mathbb{Z}[X]/(\Phi_p(X)) \to \mathbb{F}_q[X]/\overline{P(X)}$ but with a larger kernel $\langle\{q, P(X)\}\rangle$.

$$\begin{array}{ccc} \mathbb{Z}[X] & \xrightarrow{\mod \Phi_p(X)} & \mathbb{Z}[X]/\Phi_p(X) \\ \Big\downarrow{\scriptstyle \mod q} & & \Big\downarrow{\scriptstyle \mod \langle q, P(X)\rangle} \\ \mathbb{F}_q[X] & \xrightarrow{\mod \overline{P(X)}} & \mathbb{F}_q[X]/\overline{P(X)} \end{array}$$

Figure 4.2: The p-th cyclotomic polynomial over $\mathbb{Z}$ and $\mathbb{F}_q$, with $\mathrm{ord}_p(q) = r < p-1$.

In both cases, we have a good control over $\sum_{l=1}^{p-1} \alpha_{(l\cdot B)}(s_A) \in \mathbb{Z}[X]/\Phi_p(X)$ and the image of this element in the cyclotomic field $\mathbb{F}_{q^r}$.

Lemma 15. *Let $p, q \in \mathbb{Z}$ be two distinct primes. The diagrams in Figure 4.1 and Figure 4.2 commute.*

Proof. If $\mathrm{ord}_p(q) = p-1$, both maps are the canonical maps with kernel $\langle\{q, \Phi_p(X)\}\rangle \subseteq \mathbb{Z}[X]$.

If $\mathrm{ord}_p(q) = r < p-1$, the kernel of the two compositions are $\langle\{q, P(X)\}\rangle$ for the map $\mathbb{Z}[X] \to \mathbb{F}_q[X] \to \mathbb{F}_q[X]/\overline{P(X)}$ and $\langle\{\Phi_p(X), q, P(X)\}\rangle$ for the map $Z[X] \to \mathbb{Z}[X]/\Phi_p(X) \to F_q[X]/\overline{P(X)}$. Since

$$q \mid \Phi_p(X) \mod P(X)$$

we have $\Phi_p(X) \in \langle\{q, P(X)\}\rangle$, thus the kernel are the same and the diagram commutes. ∎

Theorem 16. *Let $p, q \in \mathbb{Z}$ be two distinct primes with $\mathrm{ord}_p(q) = r$. Let $F_p \in \mathbb{F}_{q^r}^{p\times p}$ be the Fourier matrix in characteristic q, see (4.1), and let $A, B \in \binom{\{0,1,\ldots,p-1\}}{k}$ be two index sets. If*

$$q \nmid \left(m_{A,B} \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \right),$$

we have $\det((F_p)_{A,B}) \neq 0 \in \mathbb{F}_{q^r}$.

Proof. Assume that $\det((F_p)_{A,B}) = 0 \in \mathbb{F}_{q^r}$, by Lemma 12 we know that $\det((F_p)_{A,l\cdot B}) = 0$ for all $l \in (\mathbb{F}_p)^\times$. Furthermore let $R_l(X) \in \mathbb{Z}[X]/\Phi_p(X)$ be the remainder of $\alpha_{l\cdot B}(s_A(x_1, \ldots, x_k))$ modulo $\Phi_p(X)$. We distinguish two cases, first $\mathrm{ord}_p(q) = p-1$ and second $\mathrm{ord}_p(q) = r < p-1$. In the first case it is $\det((F_p)_{A,l\cdot B}) = 0$ if and only if q divides every coefficient of R_l for every l. Hence we also know that q divides the coefficients of the sum

$$\sum_{l=1}^{p-1} R_l(X) = \left(m \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \right),$$

where the equality follows from Lemma 13. This is a contradiction to our assumption and hence $\det((F_p)_{A,B}) \neq 0$.

In the second case we have $\det((F_p)_{A,B}) = 0$ if and only if $R_l = 0 \mod \{\langle q, P(X)\}\rangle$ and consequently also

$$\sum_{l=1}^{p-1} R_l(X) = \left(m \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \right) = 0 \mod \langle q, P(X)\rangle.$$

Since $|m_{A,B} \cdot p - s_A(1,\ldots,1)|$ is an integer that clearly is not contained in the ideal $\langle q\rangle$ by the condition, it will be mapped to a nonzero element of degree zero in $\mathbb{F}_q[X]$. Thus it is not an element of the kernel $\langle \overline{P(X)}\rangle$. ∎

Our main result now follows directly.

Corollary 17. *Let $p, q \in \mathbb{Z}$ be two distinct primes with $\mathrm{ord}_p(q) = r$. Let $F_p \in \mathbb{F}_{q^r}^{p\times p}$ be the Fourier matrix in characteristic q, see* (4.1)*. If*

$$q > \max_{A,B\subseteq\{0,1,\ldots,p-1\},\#A=\#B} \left| m_{A,B} \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \right|,$$

then all minors of F_p are nonzero.

Proof. By Theorem 16 we need to show that

$$q \nmid \left(m_{A,B} \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \right),$$

for all $A, B \subseteq \{0, 1, \ldots, p-1\}$, with $\#A = \#B$. This is trivially the case under the condition on q and the statement follows. ∎

Example 18. *For $p = 7$ we have*

$$\max_{A,B\subseteq\{0,1,\ldots,p-1\},\#A=\#B} \left| m_{A,B} \cdot p - \frac{\prod_{i<j}(a_j - a_i)}{\prod_{i<j}(j-i)} \right| = 8$$

and thus Corollary 17 yields that the Fourier matrix F_7 has nonvanishing minors in characteristic q for every prime $q > 8$. In comparison, for $p = 7$ [17, Theorem A] yields this property of F_7 in characteristic q for any prime $q \geq 75$ and $\mathrm{ord}_7(q) = 6$.

p	2	3	5	7	11	13
q via Corollary 17	3	2	7	11	191	1619
q via [4]	3	2	11	79	105 863	11 709 007
q via [17]	3	2	13	89	105 871	11 709 007

Table 4.1: First prime q for which F_p has nonvanishing minors in characteristic q according to Corollary 17, the recent improvement [4] of [17], and [17, Theorem A].

Example 19. *For $p = 5$ and $q = 11$, we have $\mathrm{ord}_5(11) = 1$ and thus already $\mathbb{F}_{11}$ contains 5-th roots of unity, e.g. $\omega_5 = 3$. According to Corollary 17, the Fourier matrix*

$$F_5 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 & 4 \\ 1 & 9 & 4 & 3 & 5 \\ 1 & 5 & 3 & 4 & 9 \\ 1 & 4 & 5 & 9 & 3 \end{pmatrix} \in \mathbb{F}_{11}^{5\times 5}$$

has no vanishing minors in characteristic 11. *This also implies that every square submatrix is invertible over* $\mathbb{R}$ *and via standard arguments [16] the uncertainty principle*

$$\|g\|_0 + \|F_5 g\|_0 \geq 6$$

for all $g \in \mathbb{R}^5$*, where* $\|\cdot\|_0$ *denotes the number of nonzero entries of a vector.*

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